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Variance-Aware Penalized Panel Models for Temporal Risk Detection from Wearable Sensor Data

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Abstract:	Background: Wearable devices generate continuous, high-resolution physiological data that offer opportunities for real-time assessment of stress, arousal, and early physiological deterioration, but existing pipelines often treat variability as nuisance noise or rely on labeled classifiers. We present a computational approach for subject-adaptive temporal risk detection that explicitly separates conditional mean and variance dynamics in high-dimensional multisubject sensor data. Results: The proposed penalized panel ARX--GARCHX model integrates subject-specific baselines, shared autoregressive dynamics, sparse multimodal covariate effects, and covariate-dependent volatility. It produces an exceedance-based risk score that estimates the conditional probability of crossing an individualized physiological threshold. Simulation experiments across stable, seasonal, transient-regime, and sustained-regime settings showed that modeling covariate-driven variance improved recovery of threshold-exceedance risk when volatility was structured. In a Wearable Stress and Affect Detection (WESAD) demonstration, the score provided an interpretable, label-free temporal summary that separated stress-associated windows more clearly than raw heart-rate summaries and remained lightweight for streaming use. Conclusions: Variance-aware penalized panel modeling provides a reproducible methodology for converting noisy wearable streams into subject-adaptive risk-state scores. It is intended for translational feature extraction, with prospective validation required before clinical decision support.
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Variance-Aware Penalized Panel Models for Temporal Risk Detection from Wearable Sensor Data

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Abstract

Background: Wearable devices generate continuous, high-resolution physiological data that offer opportunities for real-time assessment of stress, arousal, and early physiological deterioration, but existing pipelines often treat variability as nuisance noise or rely on labeled classifiers. We present a computational approach for subject-adaptive temporal risk detection that explicitly separates conditional mean and variance dynamics in high-dimensional multisubject sensor data.

Results: The proposed penalized panel ARX–GARCHX model integrates subject-specific baselines, shared autoregressive dynamics, sparse multimodal covariate effects, and covariate-dependent volatility. It produces an exceedance-based risk score that estimates the conditional probability of crossing an individualized physiological threshold. Simulation experiments across stable, seasonal, transient-regime, and sustained-regime settings showed that modeling covariate-driven variance improves recovery of threshold-exceedance risk when volatility is structured. In the Wearable Stress and Affect Detection (WESAD) demonstration, the score provided an interpretable, label-free temporal summary that separated stress-associated windows more clearly than raw heart-rate summaries and remained lightweight for streaming use.

Conclusions: Variance-aware penalized panel modeling provides a reproducible methodology for converting noisy wearable streams into subject-adaptive risk-state scores. It is intended for translational feature extraction, with prospective validation required before clinical decision support.

Keywords: wearable sensors, digital biomarkers, temporal risk detection, heteroscedasticity, panel time series, GARCHX, physiological monitoring

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1 Background

Wearable sensing systems generate continuous multisubject, multimodal physiological streams, including heart rate, electrocardiography (ECG), electrodermal activity (EDA), respiration, body temperature, electromyography (EMG), and acceleration. These signals are increasingly used to study stress, autonomic regulation, sleep-wake patterns, activity, and broader psychophysiological states in settings that are closer to daily life than traditional clinic-based assessment [1–4]. For translational medicine, such data may provide scalable, low-burden measurements that complement clinical assessments and patient-reported outcomes.

A major obstacle is that wearable physiological signals are not simply noisy observations of stable latent states. Their temporal structure is affected by motion artifacts, device placement, context, individual baseline differences, and heteroscedastic variability. Much recent work has focused on supervised machine-learning and deep-learning models for wearable affect recognition and physiological state detection [5–10]. Related signal-processing work has emphasized denoising, adaptive filtering, instantaneous-frequency estimation, forecasting, and anomaly detection for wearable or biomedical sensor streams [11–17]. These approaches demonstrate the value of continuous physiological monitoring, but many rely on dense labels, short-window classification, or feature extraction without explicitly separating changes in location from changes in volatility. Distinguishing changes in location from changes in volatility is important in biomedical and translational studies, where a risk-relevant physiological deviation may depend on a participant's own baseline and variance pattern rather than on a fixed population threshold [18–20].

Model-based temporal analysis provides a complementary route. Autoregressive models with exogenous covariates (ARX) are useful for dynamic input-output relationships [21], and generalized autoregressive conditional heteroscedasticity models with covariates (GARCHX) allow the conditional variance to depend on lagged variation and external predictors [22–26]. However, standard implementations are usually designed for single series and low-dimensional inputs. Wearable studies, in contrast, often involve repeated measurements from multiple participants and many correlated multimodal features. A translationally useful approach should therefore pool information across participants, retain subject-specific baselines, select informative covariates, and return a risk score that can be interpreted on a probability scale. Dynamic panel modeling [27] and sparse penalization [28–31] provide useful building blocks, but they are rarely integrated with covariate-driven volatility modeling for wearable physiological risk detection. Related work on variance-guided regression for heteroscedastic data has shown that explicitly leveraging conditional variance structure can improve prediction [32]. In many sensing applications, the goal extends beyond forecasting to deriving a *temporal risk score* that captures mean dynamics and variance fluctuations under realistic noise. Wearable streams often show sharp volatility changes driven by motion or context, motivating explicit variance modeling when extracting stable temporal representations from noisy signals.

We propose a variance-aware penalized panel model for temporal risk detection from wearable sensor data. The model combines a panel ARX mean equation with a

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GARCHX variance equation, uses sparsity penalties to handle high-dimensional multimodal covariates, and outputs an exceedance-based risk score. The score estimates the conditional probability that a target physiological signal exceeds a subject-specific threshold after accounting for recent history, covariates, and time-varying variance. Figure 1 summarizes the workflow. Panel (A) shows how multimodal wearable streams are processed through the penalized ARX–GARCHX framework to obtain variance-aware temporal representations, while Panel (B) illustrates pronounced volatility changes in windowed heart-rate data from the Wearable Stress and Affect Detection (WESAD) dataset [4]; here `HR_mean` denotes the mean heart rate (in beats per minute) within each 60-s window, the target signal used throughout.

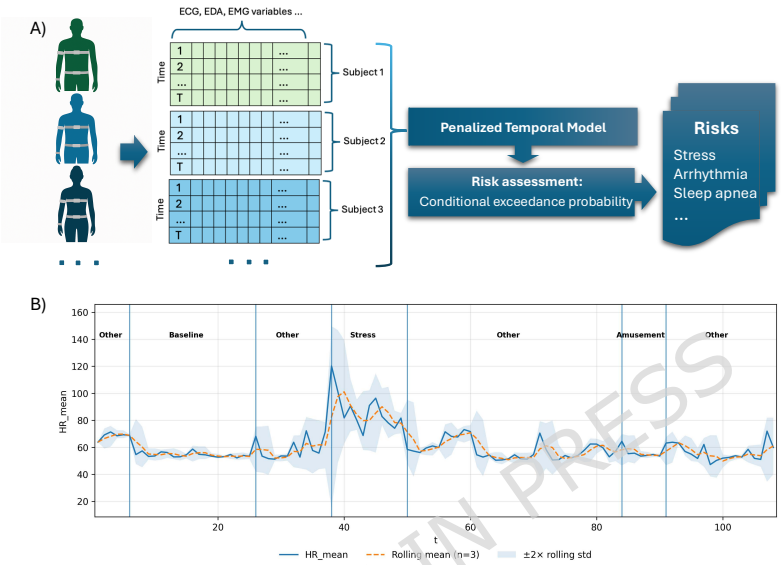


Fig. 1: Conceptual workflow for temporal risk detection from wearable sensor data. Multisubject physiological streams are represented as window-level features, analyzed with a penalized temporal model, and summarized through a conditional exceedance probability that can be used as a subject-adaptive risk assessment. Any downstream conditions shown in Panel (A), such as arrhythmia or sleep apnea, are *hypothetical* illustrative applications and are not analyzed in this study; the only empirical demonstration here is the WESAD stress benchmark. The lower panel illustrates heart-rate dynamics across annotated WESAD segments.

The aims of this study were: (i) to develop a penalized panel ARX–GARCHX model for high-dimensional multisubject wearable streams; (ii) to define an interpretable exceedance-based risk score; (iii) to evaluate recovery of the risk score in simulations with structured volatility; and (iv) to demonstrate proof-of-concept performance for stress-associated temporal risk detection using the WESAD benchmark dataset [4, 33].

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2 Methods

2.1 Penalized panel ARX–GARCHX model

Let $s = 1, \ldots, S$ index subjects and $t = 1, \ldots, T$ index time. All subjects share the same time length T, so we consider a balanced panel. We observe $\{(y_{s,t}, x_{s,t})\}$, where $y_{s,t}$ is a univariate sensor-derived signal of interest (e.g., heart rate), and $x_{s,t} \in \mathbb{R}^d$ denotes a high-dimensional vector of multimodal covariates (e.g., engineered features or concurrent sensor channels). For the response variable $y_{s,t}$, we consider an autoregressive panel data model [27] of the form

$$y_{s,t} = \alpha_s + \sum_{i=1}^{P} \theta_i \, y_{s,t-i} + \sum_{j=0}^{Q} x_{s,t-j}^{\top} \beta_j + \sigma_{s,t} \, \varepsilon_{s,t}, \tag{1}$$

where $\mathbb{E}[\varepsilon_{s,t} \mid x_{s,t}, \mathcal{F}_{t-1}] = 0$, $\mathrm{Var}(\varepsilon_{s,t} \mid x_{s,t}, \mathcal{F}_{t-1}) = 1$, and $\mathcal{F}_{t-1}$ denotes the information set generated by all past observations and covariates up to time $t-1$. The innovations $\varepsilon_{s,t}$ thus have zero conditional mean and unit conditional variance given the past, making $\sigma_{s,t}$ the conditional standard deviation. Equivalently, define $\mu_{s,t} := \alpha_s + \sum_{i=1}^{P} \theta_i \, y_{s,t-i} + \sum_{j=0}^{Q} x_{s,t-j}^{\top} \beta_j$ so that $y_{s,t} = \mu_{s,t} + \sigma_{s,t} \varepsilon_{s,t}$.

The conditional variance model is

$$\sigma_{s,t}^2 = \omega_s + \sum_{\ell=1}^{L} b_\ell \sigma_{s,t-\ell}^2 + \sum_{r=1}^{R} a_r e_{s,t-r}^2 + \sum_{m=0}^{M} x_{s,t-m}^{\top} \gamma_m, \tag{2}$$

where $e_{s,t} = y_{s,t} - \mu_{s,t}$ and ω_s is a subject-specific baseline variance. To ensure $\sigma_{s,t}^2 > 0$, we (i) enforce $a_r, b_\ell \geq 0$, $\omega_s \geq \omega_{\min} > 0$, and (ii) project the linear predictor to a positive set when needed. Implementation safeguards are detailed in Supplementary Section S2.

Sharing vs. individual effects.

Intercepts α_s (mean) and ω_s (baseline variance) are *subject-specific*, while $\{\theta_i\}, \{\beta_j\}, \{\gamma_m\}, \{a_r\}, \{b_\ell\}$ are *shared* population parameters.

2.2 Penalized quasi-likelihood

Parameters are estimated by penalized Gaussian quasi-likelihood [34]:

$$\min_{\alpha, \theta, \beta, \omega, a, b, \gamma} \sum_{s=1}^{S} \sum_{t=1}^{T} \left\{ \log \sigma_{s,t}^2 + \frac{(y_{s,t} - \mu_{s,t})^2}{\sigma_{s,t}^2} \right\} + P_\lambda(\beta) + P_\rho(\gamma), \tag{3}$$

where P_λ and P_ρ may be the ℓ_1, elastic net, smoothly clipped absolute deviation (SCAD) [35], minimax concave penalty (MCP) [36], or group penalties. In this paper, we employ the ℓ_1 penalty to select informative covariates in both the conditional mean and conditional variance equations. Model fitting uses alternating updates of the mean and variance blocks.

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Although the focus of this work is methodological and applied, the proposed penalized estimator enjoys standard high-dimensional guarantees. Under regularity, restricted-strong-convexity, and score-bound conditions on the panel process (Assumptions S1–S3 in the Supplementary Material), the penalized ARX–GARCHX estimator $(\widehat{\beta}, \widehat{\gamma})$ attains an excess-risk rate of order $(s_\beta + s_\gamma) \log p/(ST)$, where (s_β, s_γ) are the sparsity levels of the oracle pair $(\beta^\star, \gamma^\star)$ and p is the ambient dimension; the formal statement and full proof are given in Supplementary Section S1 (Theorem S1).

2.3 Algorithm: alternating penalized estimation

Given current conditional variance estimates, the subject-specific intercepts in the conditional mean equation are updated by weighted least squares, and the shared mean coefficients (θ, β) are obtained from a weighted ℓ_1-penalized regression. The variance block uses a decoupled update: (ω, a, b) is refit by constrained nonlinear optimization with γ held fixed, after which γ is updated by a proximal-gradient step under positivity safeguards. A forward recursion then refreshes $\sigma^2_{s,t}$ for the next outer iteration. The complete pseudocode (Algorithm S1), positivity enforcement (Remark on the variance recursion), and convergence/tuning details are given in Supplementary Section S2. For all experiments, covariates were standardized within each subject and the lag orders were fixed to $(P, Q, L, R, M) = (1, 0, 1, 1, 0)$, reflecting the short-memory structure typical of wearable sensor data; penalties (λ, ρ) were selected by blockwise time-series cross-validation that preserves temporal ordering within each subject.

2.4 Exceedance-based risk score

The fitted model provides estimated conditional mean $\widehat{\mu}_{s,t}$ and conditional standard deviation $\widehat{\sigma}_{s,t}$. For a subject-specific threshold c_s, we define the temporal risk score as

$$\pi_{s,t}(c_s) = \Pr(Y_{s,t} > c_s \mid \mathcal{F}_{t-1}, x_{s,t}) = 1 - F_\varepsilon \left(\frac{c_s - \mu_{s,t}}{\sigma_{s,t}} \right), \tag{4}$$

where F_ε is the innovation distribution function. We used the Gaussian distribution for the primary analysis; a standardized Student-t distribution can be used for heavier-tailed signals. Choosing c_s as a participant-specific quantile of the observed signal makes the resulting score comparable across participants while preserving subject-specific baselines.

The score is not a diagnosis. It is an interpretable physiological risk-state indicator: high values indicate that, given the recent history and multimodal covariates, the target signal is likely to exceed an individualized threshold. The score and its interpretation are explicitly conditional on the choice of threshold c_s: $\pi_{s,t}(c_s)$ answers "how likely is the signal to exceed c_s right now," so a different c_s defines a different exceedance event and a different score. The threshold should therefore be chosen to match the intended use—for example a within-subject quantile for baseline-relative monitoring, or a clinically anchored value when one is available—and reported alongside the score. We return to this dependence, and to how c_s is set in practice, in Sections 4.2 and 5.

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3 Simulation Studies

3.1 Data generating scenarios

We evaluated the proposed variance-aware autoregressive model on synthetic panel datasets designed to challenge both mean tracking and variance modeling under smooth and piecewise changes. For subjects $s = 1, \ldots, S$ and times $t = 1, \ldots, T$, we generated innovations $\varepsilon_{s,t}$ that were conditionally independent over time given $\mathcal{F}_{t-1}$ and independent across subjects. We considered either Gaussian or standardized Student-t innovations,

$$\varepsilon_{s,t} \sim \begin{cases} N(0,1), & \text{Gaussian}, \\ t_\nu / \sqrt{\nu/(\nu-2)}, & \text{Student-}t\ (\nu > 2), \end{cases} \qquad \mathrm{Var}(\varepsilon_{s,t}) = 1. \tag{5}$$

The exogenous vector $X_{s,t}$ combined a low-dimensional signal component, a binary regime indicator that induced heteroscedasticity, and a high-dimensional block of nuisance noise:

$$X_{s,t} = \left(Z_{s,t}^\top, X_{s,t}^{\mathrm{bin}}, W_{s,t}^\top\right)^\top, \quad Z_{s,t} \in \mathbb{R}^2, \quad X_{s,t}^{\mathrm{bin}} \in \{0,1\}, \quad W_{s,t} \in \mathbb{R}^{d_{\mathrm{noise}}}. \tag{6}$$

The two-dimensional signal block $Z_{s,t}$ carried the true signal. As shown in Table 1, we considered independent bivariate standard normal draws, coordinate-wise autoregressive processes with mild persistence, and a seasonal sine-cosine pair. These generators represent independent baseline variation, short-memory persistence typical of sensor drift, and diurnal or circadian-like variation. The binary regime indicator $X_{s,t}^{\mathrm{bin}}$ modulated volatility and was generated either by a within-subject median trigger or by piecewise-constant segments. The median trigger captures personalized volatility bursts, whereas the piecewise design creates sustained regime shifts resembling activity or context changes. The nuisance block $W_{s,t}$ followed a weakly persistent autoregressive process,

$$W_{s,t}^{(\ell)} = \rho W_{s,t-1}^{(\ell)} + \xi_{s,t}^{(\ell)}, \qquad \xi_{s,t}^{(\ell)} \overset{i.i.d.}{\sim} N(0,1), \qquad W_{s,0}^{(\ell)} \sim N\left(0, \frac{1}{1-\rho^2}\right), \tag{7}$$

for $\ell = 1, \ldots, d_{\mathrm{noise}}$. Modeling nuisance covariates as autocorrelated rather than white noise makes separation of true signals from background variation more realistic.

As shown in Figure 2, in Scenario S1 the ribbon varies only modestly and $y_{s,t}$ tracks $\mu_{s,t}$ closely, indicating relatively stable variance without regime-driven shifts. In Scenario S2, sinusoidal covariates generate a smooth diurnal modulation of $\mu_{s,t}$ while ribbon width remains approximately constant, so variability is primarily mean-driven. Scenario S3 has numerous short shaded intervals that coincide with brief increases in $\sigma_{s,t}$: the ribbon widens but $\mu_{s,t}$ remains level, demonstrating transient and variance-driven episodes. In Scenario S4, extended shaded blocks coincide with sustained increases in $\sigma_{s,t}$ and more frequent large deviations, consistent with heavy-tailed innovations.

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Table 1: Simulation design for Scenarios S1–S4. All scenarios share $\alpha_s \sim \mathcal{N}(0, 0.5^2)$, and the noise generator W follows the AR(1) process in Equation (7).

	S1	**S2**	**S3**	**S4**
ω_s^*	0.10	0.10	0.01	0.01
θ_i	0.40	0.50	0.40	0.40
(a_r, b_ℓ)	(0.05, 0.50)	(0.06, 0.50)	(0.06, 0.20)	(0.06, 0.20)
β_j on Z	(0.40, 0)	(0.80, 0.60)	(0.20, 0.10)	(0.80, 0.60)
γ_m on Z or X^{bin}	Z: (0.20, 0)	Z: (0.10, 0)	X^{bin}: 0.40	X^{bin}: 0.50
Innovations $\varepsilon_{s,t}$	$\mathcal{N}(0,1)$	$\mathcal{N}(0,1)$	$\mathcal{N}(0,1)$	t_ν (ν=8)
Signal Z	i.i.d. $\mathcal{N}_2(0, I_2)$	seasonal $(\sin, \cos)^{\S}$	AR(1)	seasonal $(\sin, \cos)^{\S}$
Regime X^{bin}	none	none	subject median†	piecewise‡
Burn-in	200	220	180	180

Note. Parameter values are held constant across indices i, r, ℓ, j, and m within each scenario. The burn-in period is used only during data generation to remove initialization transients and is discarded before model fitting. Consequently, it does not affect the estimation procedure, the estimands, or the reported Bias and RMSE. The modest differences in burn-in lengths across scenarios were chosen for simulation convenience to ensure that the retained observations are approximately from the stationary regime.

$*$ ω_s is set to be the same for all subjects s.

§ $Z^{(1)} = \sin(2\pi t/T_P)$ and $Z^{(2)} = \cos(2\pi t/T_P)$ are seasonal covariates shown in Figure 2(B,D).

† Subject-median trigger $X_{s,t}^{\text{bin}} = \mathbb{1}\{Z_{s,t}^{(1)} > \text{median}_t(Z_{s,t}^{(1)})\}$, computed per subject over the post-burn-in window; this produces frequent, short high-variance bursts.

‡ Piecewise-constant $X_{s,t}^{\text{bin}}$ alternates between 0 and 1, with run lengths jittered around a target scale T_L shared across subjects, yielding sparse, long-duration blocks.

3.2 Competing models and evaluation metrics

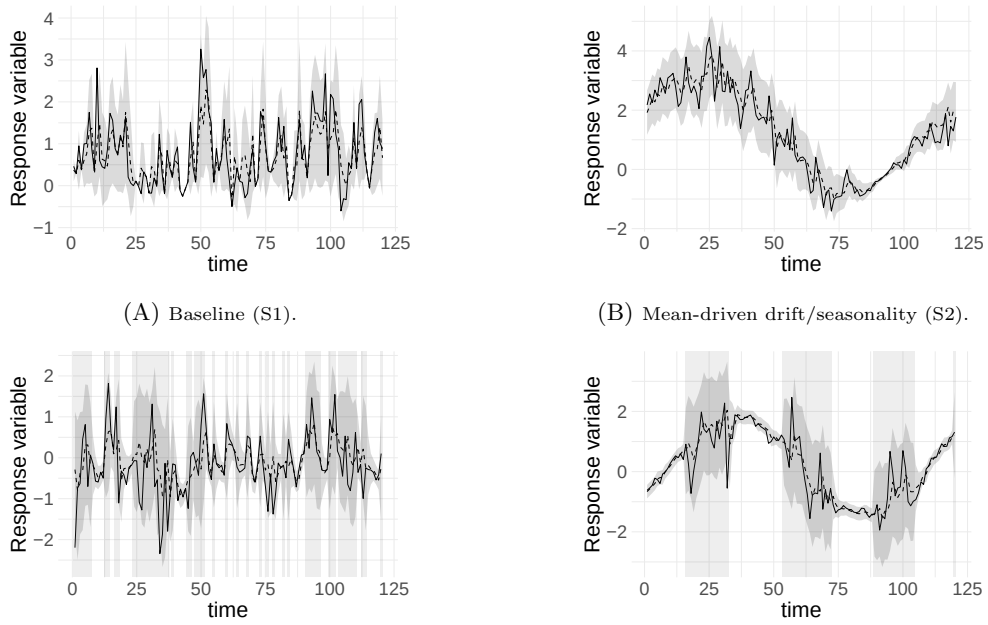
We compare six variants within the ARX–GARCH family (Models A–F). Models A–D include covariates in the conditional mean and differ in whether the mean or variance components are penalized and whether X also enters the variance equation; Model E adds the intermediate specification with a penalized mean and an *unpenalized* X-dependent variance, and Model F is a covariate-free AR–GARCH baseline:

- **Model A**: ARX–GARCH (variance independent of X),
- **Model B**: ARX$_p$–GARCH (penalized mean; variance independent of X),
- **Model C**: ARX–GARCHX (variance includes X, unpenalized),
- **Model D**: ARX$_p$–GARCHX$_p$ (penalized mean and penalized variance),
- **Model E**: ARX$_p$–GARCHX (penalized mean; unpenalized X-dependent variance),
- **Model F**: AR–GARCH (no covariates in the mean or variance; baseline).

Model E isolates the effect of penalizing the variance covariates (E versus D) while holding the penalized mean fixed, and Model F quantifies the value added by covariates over a purely autoregressive, covariate-free specification. Here ARX$_p$ and GARCHX$_p$ denote the ARX mean and GARCHX variance equations with ℓ_1 penalties applied to the covariate coefficients β and γ, respectively.

To focus on relevant tails, we evaluate cutoffs c at pooled empirical quantiles $p \in \{0.60, 0.70, 0.80, 0.90, 0.95\}$, providing global thresholds that preserve between-subject heterogeneity at small T. We fix lag orders $(P, Q, L, R, M) = (1, 0, 1, 1, 0)$ with

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(A) Baseline (S1). (B) Mean-driven drift/seasonality (S2).

(C) Regime contrast (S3). (D) Seasonal + piecewise regime (S4).

Fig. 2: Illustration of one example subject for each scenario (S1–S4) with $y_{s,t}$ (solid), $\mu_{s,t}$ (dashed), and the band $\mu_{s,t} \pm 2\sigma_{s,t}$. For Scenarios 3 and 4, gray shading denotes intervals where the regime indicator is active (X^{bin}=1).

standardized predictors. From each fitted model, we obtain $(\widehat{\mu}_{s,t}, \widehat{\sigma}_{s,t})$ and compute $\widehat{\pi}_{s,t}(c)$. For a given threshold c, we compare the estimate $\widehat{\pi}_{s,t}(c)$ with the oracle $\pi_{s,t}(c)$ via

$$\text{Bias}(c) = \frac{1}{ST}\sum_{s=1}^{S}\sum_{t=1}^{T}\big(\widehat{\pi}_{s,t}(c) - \pi_{s,t}(c)\big), \text{ and } \text{RMSE}(c) = \sqrt{\frac{1}{ST}\sum_{s=1}^{S}\sum_{t=1}^{T}\big(\widehat{\pi}_{s,t}(c) - \pi_{s,t}(c)\big)^2}.$$

We averaged the bias and RMSE across the five cutoff values to summarize each replication. Simulations were conducted with $S = 15$ subjects, $d_{\text{noise}} = 100$ noise covariates, and time horizons $T \in \{120, 240\}$, chosen to match the scale of our real-data application to the WESAD study. We set $\rho = 0.5$ for the AR(1) noise variables W and $\rho = 0.6$ for the signal variable Z in Scenario 3. In Scenarios 2 and 4, we used a seasonal period $T_P = T/3$ for the covariate Z, and Scenario 4 additionally used a piecewise regime run length of $T_L = 20$ for the binary covariate X^{bin}. Each scenario was replicated 100 times.

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3.3 Simulation results

Across all simulated settings, the accuracy of the exceedance-based risk score is governed by whether the conditional variance is allowed to depend on covariates and whether that dependence matches the structure of the underlying volatility. When volatility is stable (Scenario 1), all six models perform similarly, so modeling covariate-driven variance costs little where it is unnecessary. Once volatility becomes structured—through seasonality (Scenario 2), regime changes (Scenario 3), or the two combined with heavy-tailed innovations (Scenario 4)—the variance-aware specifications separate clearly from the variance-agnostic baselines, attaining lower RMSE and near-zero bias in the risk score (Figure 3). The proposed specification **D** is strongest in the seasonal, wearable-like regime (Scenario 2), and its margin widens as the regime structure intensifies (Scenarios 3–4) and as the horizon lengthens to T=240; even under the heavy-tailed innovations of Scenario 4, **C** and **D** remain robust and continue to outperform the variance-agnostic baselines. Extending the comparison to all six models refines rather than overturns this picture. Model **E** (ARX$_p$–GARCHX; penalized mean, unpenalized X-dependent variance) is essentially indistinguishable from **D**—indeed the most accurate model at the longer horizon T=240, with the lowest RMSE in all four scenarios—indicating that penalizing the variance covariates chiefly stabilizes estimation at small T rather than shifting the central tendency. At the other extreme, the covariate-free baseline **F** (AR–GARCH) has by far the highest RMSE in every scenario (roughly 0.10–0.16 versus 0.05–0.09 for the covariate-driven models), quantifying the accuracy lost by ignoring exogenous covariates altogether. Taken together, the six models bracket the value of covariate-driven variance: decisive when volatility is structured, negligible when it is not, and best realized by the penalized specification that couples covariate information with regularized estimation. Full six-model results are reported in Supplementary Table S1.

Both variance-aware models **C** and **D** exhibit a mild positive bias, reflecting slight overestimation of exceedance probabilities, but the mechanism differs between them. In Model **D**, which applies an ℓ_1 penalty to the variance-covariate coefficients γ, the bias reflects shrinkage of γ toward zero, which mildly attenuates the fitted conditional variance and shifts exceedance probabilities upward through the curvature of $z \mapsto 1 - \Phi(z)$. Model **C**, by contrast, applies *no* penalty in the variance equation; its bias instead arises from finite-sample estimation noise when the high-dimensional variance-covariate vector is fit without regularization (many near-zero coefficients estimated from short series), together with the positive-variance safeguard, which together inflate small fitted variances. In both cases the effect is small. Even so, their RMSE remains lowest overall, indicating that the variance-aware formulations achieve a favorable bias–variance tradeoff and provide the most reliable exceedance-based temporal summaries across scenarios and time horizons. Replication-level Bias and RMSE means with standard errors are tabulated in Supplementary Table S1.

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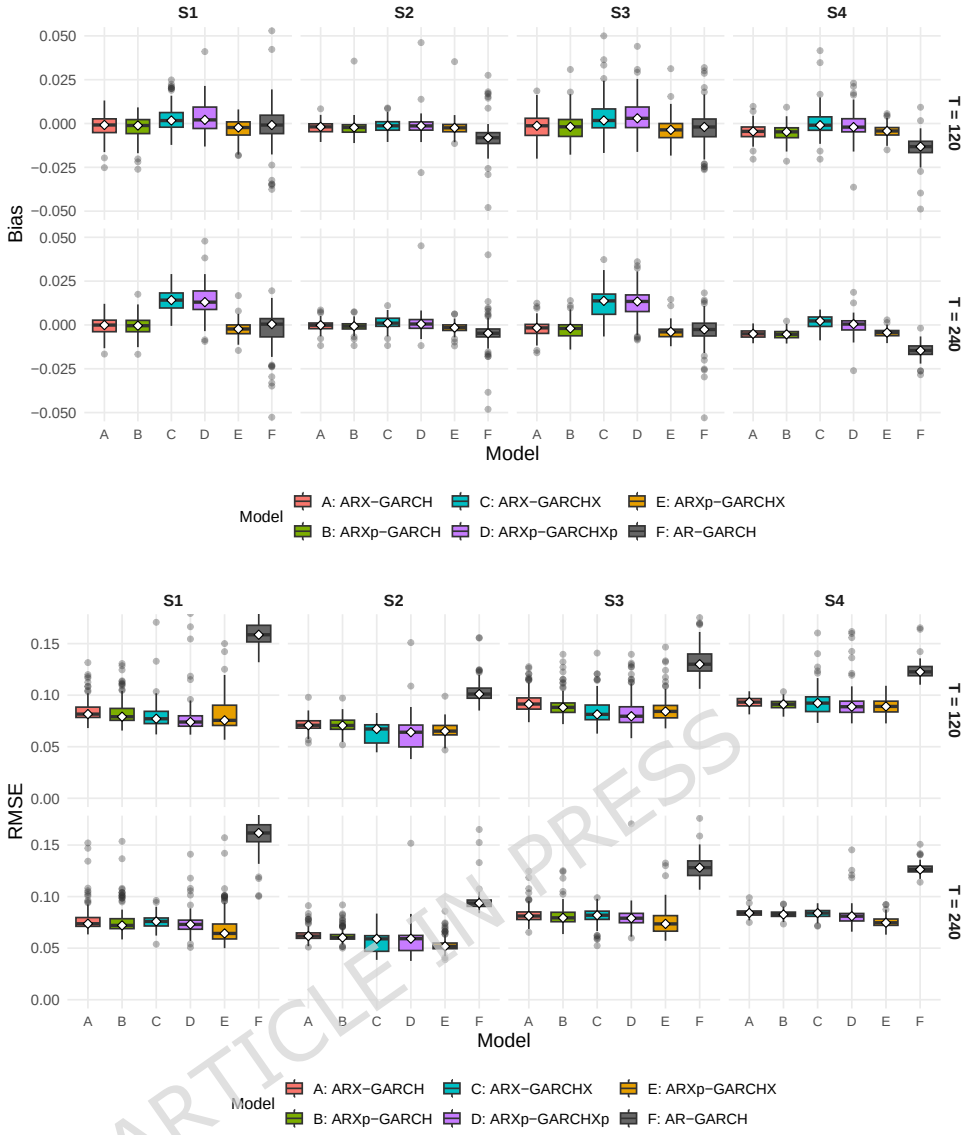


Fig. 3: Bias (top) and RMSE (bottom) of the estimated risk score across simulation Scenarios S1–S4 and Models A–F, at horizons $T = 120$ and $T = 240$ (boxplots over 100 replications; the white diamond marks the median; the legend identifies the six model variants). Model A: ARX–GARCH; Model B: ARX$_p$–GARCH; Model C: ARX–GARCHX; Model D: ARX$_p$–GARCHX$_p$; Model E: ARX$_p$–GARCHX; Model F: AR–GARCH (covariate-free baseline). The covariate-free baseline (F) has by far the highest RMSE in every setting, and the variance-aware models (C, D, E) are the most accurate, with E and D essentially indistinguishable.

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4 The WESAD Lab Study

We evaluate the proposed framework using the WESAD dataset [4, 33], a controlled multimodal wearable-sensing benchmark involving 15 adult participants. Each subject completed a scripted session consisting of several annotated segments (e.g., resting, task-engaged, and amusement periods), providing clear temporal structure for analysis. Participants wore two devices—a chest-worn RespiBAN and a wrist-worn Empatica E4—so all physiological streams are time-synchronized across modalities.

The chest device provides ECG, EDA, EMG, respiration, skin temperature, and 3-axis acceleration, while the wrist device provides photoplethysmography (PPG/BVP), EDA, temperature, and acceleration. Because the protocol follows a predefined sequence of segments and the recordings are concurrent across modalities, WESAD offers clean annotations and consistent multisensor coverage suitable for multisubject or subject-wise analysis.

In our study, we focus on the chest-worn streams, as they provide the most precise heart-rate estimation. Heart rate serves as the target signal for temporal modeling due to its continuity, stability under windowing, and sensitivity to motion and other factors that drive heteroscedastic behavior in wearable data. The remaining modalities (EDA, EMG, RESP, ACC, TEMP) are incorporated as covariates to inform the covariate-driven mean and variance dynamics within the proposed variance-aware temporal risk model. The dataset annotations are used only for visualization and reference-based evaluation, not for model fitting.

4.1 Data preparation

Preprocessing and feature extraction were performed in Python using toolboxes designed for physiological-signal analysis. NeuroKit2 was used for ECG/PPG peak detection and for deriving HR, heart-rate variability (HRV), respiration, and EDA features [37]. For chest-worn devices, all streams were segmented into fixed 60-s windows with a 60-s hop. A window was retained only if at least 80% of its samples were finite across the required channels. To ensure temporal alignment across modalities, the raw streams were anti-aliased and resampled onto a common low-rate timeline prior to computing windowed features and cross-modal correlations.

Feature construction was carried out per 60-s window. From the synchronized multimodal series, we computed (i) per-series moments (mean, standard deviation, skewness, and excess kurtosis) and (ii) all pairwise Pearson cross-modal correlations to capture coordinated multimodal changes, a common pattern in wearable sensing where channels co-modulate under shared motion or environmental drivers. This produced a larger set of candidate window-level features, which were reduced to $d = 77$ by removing one member of each pair of highly collinear features (absolute Pearson correlation above 0.90, using `caret::findCorrelation`); the retained 77 features enter the model and are the covariates listed in Table 2. Windows with fewer than 80% finite samples in a required channel were dropped before feature construction, and any window with a non-finite feature value after construction was excluded; no imputation was used. After these steps, 1,427 windows remained across the 15 participants (87–117 windows per participant; participants S1 and S12 are excluded following the

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WESAD convention). The complete list of the 77 features is given in Supplementary Section S8 (Supplementary Table S6), and the retained window count for each participant is given in Supplementary Table S3. Because participants contribute different numbers of windows, the analyzed panel is in fact *unbalanced*; the balanced-panel notation of Section 2.1 is used only for expositional clarity, and the estimator constructs all lags within each participant separately (see Supplementary Section S2), so unequal series lengths are handled without truncation or padding.

For reference-based evaluation, each window was also assigned a dataset annotation (baseline, stress, amusement, or transition) using majority vote over the WESAD labels. These annotations were not used when estimating the variance-aware panel model but served only for visualization and for assessing whether the derived exceedance-based risk score provides clearer temporal differentiation than the raw heart-rate values.

Table 2: Chest-worn multimodal features used in the models (60 s windows). Summaries are grouped by sensing block.

Block	Summary of features
ECG / HRV	Short-term variability (RMSSD/pNNx), LF/HF power and ratio, simple Poincare geometry, and compact entropy/complexity indices
Respiration & BRV	Breathing rate and amplitude (levels and variability), inspiratory/expiratory durations, duty cycle, and breath-to-breath variability
EDA	Tonic level (mean/dispersion/trend) and phasic activity (SCR count, typical amplitude, cumulative phasic area)
EMG	Broadband power, median frequency, spectral entropy, and zero-crossing rate
Accelerometry	Movement intensity (RMS) and dynamics (jerk), axis medians, high-activity tail (P90), plus overall SD/skewness/kurtosis
Skin temperature	Level, per-minute slope (trend), and detrended variability
Cross-channel correlations	Short-lag correlations among RESP, EDA, ACC, TEMP, and EMG to capture coordinated multimodal changes

Note. RMSSD: root mean square of successive differences; pNNx: percentage of successive NN intervals differing by more than x ms; LF/HF: low-/high-frequency power ratio; BRV: breathing rate variability; SCR: skin conductance response; RMS: root mean square; SD: standard deviation; RESP: respiration; ACC: accelerometer; TEMP: skin temperature.

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4.2 Model specification and analysis plan

We model the 60-s windowed heart-rate mean (`HR_mean`), denoted $y_{s,t}$, as a stable, continuous signal that captures temporal fluctuations in wearable data. An ARX–GARCHX model with subject-specific intercepts and contemporaneous covariates in both the mean and variance components is fitted according to Equations (1) and (2). The lag orders are fixed at $(P, Q, L, R, M) = (1, 0, 1, 1, 0)$, a structure suitable for short-memory wearable streams, and ℓ_1 regularization is applied to (β, γ) to promote parsimony and mitigate collinearity among the multimodal features.

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To provide a normalized exceedance-based risk score across subjects, we evaluate the conditional exceedance probability (4) at subject-specific thresholds. For a chosen quantile p, we set $c_s = c_s(p) := \text{Quantile}_p(\{y_{s,t}\}_t)$ – the p-th quantile of subject s's observed signal – and substitute this c_s into (4) under the assumed innovation distribution (Gaussian or standardized t). For the WESAD analysis we set $p = 0.70$, so that $c_s(0.70)$ is the 70th percentile of each participant's observed `HR_mean` series; this individualized threshold makes the score comparable across participants while preserving subject-specific baselines. The resulting exceedance probabilities summarize the joint mean–variance dynamics in a compact, interpretable form that is independent of downstream labels.

To clarify the interpretation and estimation of the proposed score, $\widehat{\pi}_{s,t}(c_s)$ denotes the estimated conditional probability that `HR_mean` exceeds the individualized threshold c_s; it is *not* a probability of stress or of any clinical event. Because the conditional variance model uses contemporaneous covariates $x_{s,t}$ (lag order M=0), $\widehat{\pi}_{s,t}$ represents the current window rather than a one-step-ahead forecast. A forecasting variant can be obtained by restricting both the mean and variance equations to strictly lagged covariates. In this demonstration, the threshold c_s, the within-subject covariate standardization, and the subject-specific parameters (α_s, ω_s) are all derived from the analyzed recordings. In a prospective deployment, these quantities would instead be obtained during an initial subject-specific calibration period and then held fixed during subsequent monitoring. Specifically, $c_s(0.70)$ is defined as the 70th percentile of each subject's `HR_mean` distribution over the full recording. It is therefore computed without using any stress annotations, although it is estimated from the entire recording, including the windows that are later used for evaluation. A sensitivity analysis that instead estimates c_s from a leak-free calibration period is reported in Supplementary Section S6 and summarized in Section 4.3.

Evaluation endpoint.

For reference-based evaluation, we define a binary window-level endpoint. The positive class consists of windows annotated as *stress* (WESAD label 2), whereas the negative class pools all other retained windows (baseline, amusement, and transition/other). Transition and amusement windows are therefore treated as non-stress rather than discarded. After preprocessing, 1,427 windows from 15 participants were available (182 stress and 1,245 non-stress); the corresponding per-participant window and class counts are reported in Supplementary Table S3. The annotations are used only for evaluation and never for model fitting.

For comparison with supervised baselines, we implemented logistic regression, random forest, and gradient boosting using standard R packages (`stats::glm` [38], `randomForest::randomForest` [39], and `gbm::gbm` [40]). The supervised classifiers were trained and evaluated under leave-one-subject-out (LOSO) cross-validation, whereas the proposed exceedance score is unsupervised and was evaluated in-sample (fitted once to each participant's full recording and scored on the same windows). Because the proposed score and the supervised classifiers are not evaluated under a common held-out design, the supervised AUCs are presented as *contextual benchmarks* that illustrate the performance of purely discriminative window-level methods,

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rather than as a formal head-to-head comparison. The intended primary comparison for the proposed score is against the raw `HR_mean` baseline, which is evaluated under the same experimental design. These ensemble methods have shown strong discriminative performance in related translational classification tasks [41], but they do not model temporal dependence or subject-level baselines, in contrast to the proposed ARX–GARCHX risk-scoring model.

4.3 Results and subject-level insights

Using subject-specific thresholds $c_s(0.70)$ and Gaussian innovations, we evaluated the exceedance-based risk score $\widehat{\pi}_{s,t}$ against the raw 60-s heart-rate means (`HR_mean`). The pooled Receiver Operating Characteristic (ROC) analysis in Figure 4(A) shows that the variance-aware risk score provides substantially stronger separation across the annotated WESAD segments. The estimated AUC for the risk score was $\mathrm{AUC}_{\mathrm{ex}} = 0.931$, compared with $\mathrm{AUC}_{\mathrm{HR}} = 0.868$ for the raw heart-rate signal ($\Delta = 0.063$). Because repeated windows are clustered within only 15 participants, the pooled DeLong test may overstate precision. We therefore additionally report participant-clustered bootstrap confidence intervals obtained by resampling whole participants (3,000 bootstrap replicates). This yields 95% confidence intervals of [0.887, 0.972] for $\mathrm{AUC}_{\mathrm{ex}}$ and [0.821, 0.918] for $\mathrm{AUC}_{\mathrm{HR}}$, while the paired AUC difference has a 95% confidence interval of [0.031, 0.094], excluding zero. The improvement of the variance-aware risk score over the raw heart-rate signal is thus robust to within-participant clustering, indicating that modeling both mean and variance dynamics yields a more discriminative temporal representation than relying on raw heart-rate measurements alone.

For reference-based benchmarking, we also compared the proposed *unsupervised* metric with several *supervised* classifiers trained on window-level features using leave-one-subject-out cross-validation. Under LOSO cross-validation, logistic regression achieved $\mathrm{AUC}_{\mathrm{logit}} = 0.931$, random forest achieved $\mathrm{AUC}_{\mathrm{RF}} = 0.921$, and gradient boosting achieved $\mathrm{AUC}_{\mathrm{GBM}} = 0.948$. These supervised models use the annotations during training and are evaluated on held-out subjects, whereas the proposed score is unsupervised and evaluated in-sample; the two are therefore not evaluated under a common design, and we present the supervised values as contextual benchmarks rather than as a formal head-to-head comparison. Read in that light, the variance-aware score—estimated without any annotations—attains discrimination in the same range as these purely discriminative window-level classifiers, supporting its ability to extract informative temporal structure from multimodal wearable streams.

Two additional analyses examine the robustness of our results and identify the factors contributing to the observed performance. First, replacing the full-recording threshold $c_s(0.70)$ with a leak-free threshold estimated solely from an early burn-in window reduces $\mathrm{AUC}_{\mathrm{ex}}$ from 0.931 to approximately 0.880; notably, this remains at or above the raw `HR_mean` baseline (details in Supplementary Section S6). Second, we conduct a focused ablation to isolate the impact of covariate-driven variance by comparing Model B (no covariates in the variance equation) to Model D (the full model); this analysis, along with a calibration assessment for the exceedance event $\{y_{s,t} > c_s(0.70)\}$, is reported in Supplementary Section S7. Both variance-aware models substantially

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outperform the raw `HR_mean` baseline and perform comparably to each other. This indicates that the real-data advantage is primarily driven by variance-aware modeling and is robust to the inclusion of covariates in the variance equation, whereas the benefits of covariate-driven variance are more pronounced in the structured-volatility simulation regimes.

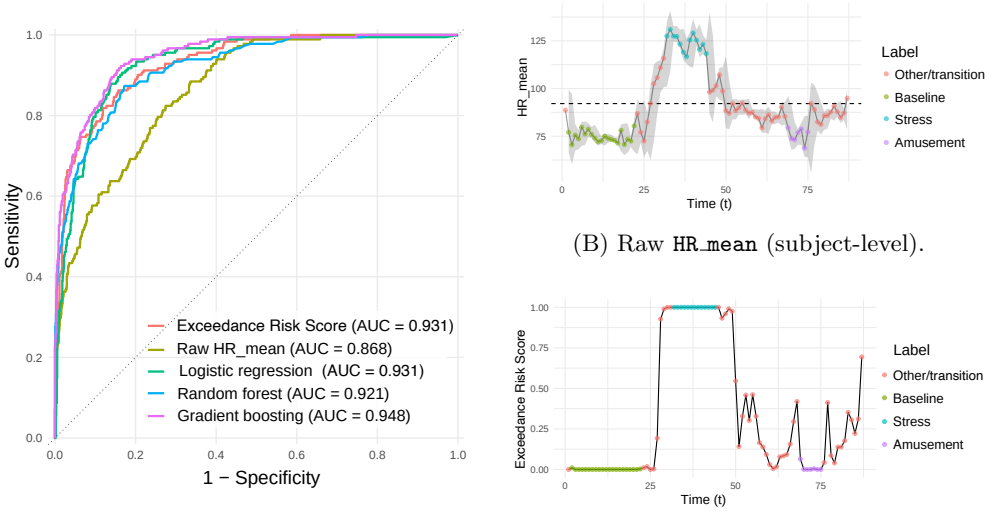
It is worth noting that the supervised baselines treat all windows as independent samples and do not incorporate subject-specific baselines or temporal dependence. They therefore serve as strong discriminative benchmarks but do not capture the autoregressive or heteroscedastic structure exploited by the proposed temporal risk-scoring framework.

To complement the aggregate ROC analysis and highlight subject-level dynamics, we examine a representative individual trajectory. Figure 4(B) shows the raw 60-s `HR_mean` series for a representative participant (WESAD subject S11), with the subject-specific 70% threshold $c_s(0.70) \approx 92.10$ indicated by a dashed line. This participant was selected before qualitative inspection of the model output as a representative case whose protocol contains a clearly delineated stress segment; trajectories for all participants are provided in the replication materials. The annotated WESAD segments reveal a sustained elevation in heart rate during one region of the protocol, with lower variability elsewhere. Occasional excursions approach the threshold outside this segment but do not persist.

Figure 4(C) displays the corresponding risk score $\widehat{\pi}_{s,t}$. The score increases sharply at the onset of the high-variability segment, remains elevated throughout its duration, and drops promptly at its conclusion. Outside this interval, the score remains near zero despite moderate fluctuations in the raw signal. This behavior indicates that the variance-aware temporal model concentrates exceedance probability in periods characterized by sustained deviations relative to the subject's own baseline while suppressing short-lived fluctuations. Together, the two right-hand panels illustrate that the exceedance formulation provides a subject-adaptive, interpretable temporal summary that aligns closely with the annotated structure of the signal.

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(A) ROC curves: pooled discrimination.

(B) Raw `HR_mean` (subject-level).

(C) Estimated risk score.

Fig. 4: WESAD results. **(A)** ROC curves comparing the variance-aware exceedance-based risk score with the raw 60-s heart-rate mean (`HR_mean`) and several supervised window-level baselines. The risk score and `HR_mean` are *unsupervised* (no annotation information used during model fitting) and evaluated in-sample, whereas logistic regression, random forest, and gradient boosting are *supervised* classifiers trained and evaluated using leave-one-subject-out cross-validation and shown as contextual benchmarks; the unsupervised risk score achieves AUC 0.931 (participant-clustered bootstrap 95% CI $[0.887, 0.972]$) and substantially improves upon the raw `HR_mean` baseline (0.868; difference 95% CI $[0.031, 0.094]$). **(B)** Subject-level raw 60-s `HR_mean` for one representative participant with the subject-specific 70% threshold $c_s(0.70) \approx 92.10$ shown as a dashed line. **(C)** The corresponding estimated risk score over time, showing elevated values during the annotated stress-associated segment and near-zero values elsewhere.

The risk score offers a principled and interpretable way to summarize temporal deviations in wearable sensor streams. Its probabilistic range between 0 and 1 provides a normalized scale that is comparable across subjects and robust to individual baseline differences. Unlike raw heart-rate thresholds, the score is sensitive to sustained departures yet resistant to transient fluctuations, yielding a stable temporal characterization of signal patterns. Importantly, the score is estimated without using any annotations, yet it naturally differentiates periods of elevated and quiescent activity in the WESAD protocol. This highlights its utility as an unsupervised temporal feature, suitable both for standalone interpretation and as a noise-robust, low-dimensional input to downstream sensing or machine-learning models.

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4.4 Computational feasibility

The WESAD analysis involved fifteen participants, approximately one hundred twenty windows per participant, and seventy-seven covariates. With twenty outer iterations, the mean-update block ran in roughly 0.01 s per iteration and the variance-update block in roughly 0.12 s per iteration, so end-to-end training completed in approximately 26.6 s. After training, a new-window score was computed with sub-millisecond latency. Detailed timings, asymptotic complexity ($O(STd)$ for the mean recursion and $O(ST(L+R+M))$ for the variance recursion), memory footprint, and hardware information are reported in Supplementary Table S2 (Supplementary Section S4). These results support potential use as a lightweight feature-extraction or monitoring component, although clinical deployment would require prospective validation.

5 Discussion

This study reframes variance-aware wearable-sensor modeling as a translational temporal risk-detection problem. The main finding is that explicitly modeling conditional variance, rather than treating variability as nuisance noise, yields an interpretable subject-adaptive score that improves discrimination of stress-associated physiological windows relative to raw heart-rate summaries. In WESAD, the risk score achieved an AUC of 0.931 without using labels during model fitting—compared with 0.868 for the raw heart-rate mean, a difference that remains significant under a participant-clustered bootstrap—while attaining discrimination in the same range as supervised logistic-regression, random-forest, and gradient-boosting classifiers evaluated as contextual benchmarks, and remaining computationally lightweight. We caution that the supervised classifiers were trained and evaluated under leave-one-subject-out cross-validation whereas the unsupervised score was evaluated in-sample, so this correspondence is contextual rather than a formal head-to-head comparison.

The method has several translationally relevant properties. First, it produces a probability-scale score between 0 and 1, which is easier to interpret than arbitrary transformed features. Second, it uses individualized thresholds, making the score more robust to baseline differences across participants. Third, it incorporates multimodal covariates into both the mean and variance equations, allowing the model to distinguish sustained physiological shifts from transient fluctuations. Fourth, it can be computed rapidly after training, which is important for real-time or near-real-time wearable monitoring.

The approach also complements supervised machine learning. Deep learning and sequence models may achieve strong classification accuracy when large labeled datasets are available [42, 43], but many translational applications lack dense, reliable labels. The proposed score can be used as an unsupervised or weakly supervised feature that summarizes subject-specific risk dynamics before downstream prediction. It may therefore be useful as a digital biomarker component in studies of stress, autonomic dysregulation, symptom exacerbation, or other temporally dynamic states. The same idea may also connect with broader time-series segmentation and change-detection methods for multivariate data [44].

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Several limitations should be emphasized. The WESAD analysis is a controlled laboratory benchmark with a small number of participants and should not be interpreted as clinical validation. Stress labels in WESAD represent protocol-defined affective states rather than clinically diagnosed stress disorders. External validation in larger and more diverse cohorts is needed before any medical claim can be made. In addition, the supervised classifiers were evaluated under leave-one-subject-out cross-validation, while the risk score was fitted without labels and evaluated post hoc; a fully prospective validation should freeze preprocessing, tuning, thresholds, and model parameters within an outer held-out-subject or external-cohort protocol. The current model also uses fixed lag orders and a balanced-panel formulation. Irregular sampling, missingness, longer-range temporal dependence, and nonlinear mean-variance effects remain important extensions.

Finally, the risk score is threshold-dependent. While we used the 70th percentile of each participant's heart-rate distribution for interpretability and cross-participant comparison, alternative thresholds may be better suited for different clinical or behavioral endpoints. Because this quantile is computed over the entire recording, it inherently incorporates the same elevated heart-rate windows that later define the positive class. Although no labels are used, this may introduce a modest optimistic bias to the reported discrimination. A sensitivity analysis fixing c_s from a leak-free early burn-in window (Supplementary Section S6) reduces $\mathrm{AUC}_{\mathrm{ex}}$ from 0.931 to approximately 0.880 but remains at or above the raw heart-rate baseline. This confirms that the temporal signal is genuine while emphasizing that the exact magnitude of performance is sensitive to threshold selection. In future translational applications, thresholds should be fixed prospectively from an initial calibration period and chosen from clinical anchors, patient-reported outcomes, or adverse-event definitions, rather than relying solely on empirical quantiles computed over the full recording.

6 Conclusions

A penalized panel ARX–GARCHX model can convert high-dimensional wearable physiological streams into an interpretable, subject-adaptive temporal risk score. In simulations, variance-aware modeling improved recovery of threshold-exceedance risk under structured volatility. In the WESAD benchmark, the score improved stress-associated window discrimination relative to raw heart rate and performed competitively with supervised baselines while requiring less than 1 ms for new-window inference after training. The method is a promising digital biomarker workflow for translational wearable studies, but prospective external validation is required before clinical use.

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List of abbreviations

ACC	acceleration
ARX	autoregressive model with exogenous covariates
AUC	area under the receiver operating characteristic curve
BRV	breathing-rate variability
ECG	electrocardiogram
EDA	electrodermal activity
EMG	electromyography
GARCHX	generalized autoregressive conditional heteroscedasticity model with exogenous covariates
HR	heart rate
HRV	heart-rate variability
QMLE	quasi-maximum-likelihood estimation
RESP	respiration
RMSE	root mean squared error
ROC	receiver operating characteristic
WESAD	Wearable Stress and Affect Detection

Supplementary information

The Supplementary Material is organized into eight sections. Supplementary Section S1 (*Theoretical Results*) states the regularity, restricted-strong-convexity, and score-bound assumptions and proves a finite-sample oracle inequality (Theorem S1) showing that the penalized estimator achieves an excess-risk rate of order $(s_\beta + s_\gamma) \log p/(ST)$. Supplementary Section S2 (*Algorithmic Details*) gives the full estimation pseudocode (Algorithm S1) for the penalized panel ARX–GARCHX model, including the weighted closed-form mean update, the decoupled (ω, a, b) and γ variance updates, positivity safeguards, and tuning conventions. Supplementary Section S3 (*Supplementary Simulation Results*) reports the replication-level Bias and RMSE summaries for Models A–F under Scenarios S1–S4 at $T \in \{120, 240\}$ in Supplementary Table S1. Supplementary Section S4 (*Supplementary Computational Results*) reports the runtime profile, asymptotic complexity, memory footprint, and hardware specification for the WESAD analysis in Supplementary Table S2. Supplementary Section S5 (*WESAD Endpoint and Per-Participant Composition*) defines the evaluation endpoint, tabulates per-participant window and class counts (Table S3), and reports participant-clustered bootstrap confidence intervals. Supplementary Section S6 (*Sensitivity to the Threshold Choice*) reports the leak-free threshold analysis. Supplementary Section S7 (*WESAD Variance-Covariate Ablation*) compares Model B and Model D on discrimination and calibration of the exceedance event. Supplementary Section S8 (*Reproducibility Details*) lists the complete 77-feature set (Table S4), retained windows, quality control, tuning grid, temporal folds, and software/code provenance.

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Declarations

Ethics approval and consent to participate

The present study was a secondary analysis of the publicly available, de-identified WESAD dataset and did not involve new recruitment, intervention, or data collection from human participants. Ethical and consent procedures for the original WESAD data collection were reported by the original dataset investigators [4]. No additional ethics approval was required for this secondary analysis.

Consent for publication

Not applicable.

Trial registration

Not applicable. This study is a methodological article that uses publicly available secondary data and does not constitute a clinical trial.

Reporting guidelines

Not applicable in the formal sense. This is a methodological article using publicly available secondary data; established clinical reporting checklists such as TRIPOD (for prediction-model development and validation) and STARD (for diagnostic accuracy) do not directly apply because the analysis was a proof-of-concept feature-extraction demonstration on a public benchmark, with no prospective enrollment, diagnostic claim, or external clinical validation. Where relevant, we have followed the spirit of the TRIPOD-AI items by reporting the data source, preprocessing pipeline, model specification, tuning procedure, evaluation metrics, intended use, and limitations.

Availability of data and materials

The WESAD dataset analyzed in this study is publicly available through the UC Irvine Machine Learning Repository and is described in the original dataset publication [4, 33].

Code availability

An R package implementing the proposed framework, `varGuidTS`, is available at https://cloud.r-project.org/web/packages/varGuidTS/. Additional source code and replication materials for the analyses reported in this paper are available at https://github.com/zionwzz/variance-guided-risk-demo.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

ZW contributed to methodology, software implementation, data preprocessing, simulation studies, real-data analysis, visualization, and manuscript drafting. ML contributed to conceptualization, methodology, supervision, funding acquisition, interpretation of results, and manuscript revision. Both authors read and approved the final manuscript.

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Use of AI-assisted tools

During preparation of this journal-specific draft, the authors used an AI language model to assist with editorial restructuring and wording. The authors reviewed and are responsible for all scientific content, analyses, interpretations, and conclusions.

References

[1] Smets E, Rios Velazquez E, Schiavone G, Chakroun I, D'Hondt E, De Raedt W, et al. Large-scale wearable data reveal digital phenotypes for daily-life stress detection. NPJ digital medicine. 2018;1(1):67.

[2] Valenza G, Nardelli M, Lanata A, Gentili C, Bertschy G, Paradiso R, et al. Wearable monitoring for mood recognition in bipolar disorder based on history-dependent long-term heart rate variability analysis. IEEE Journal of Biomedical and Health Informatics. 2013;18(5):1625–1635.

[3] Stan IE, D'Auria D, Napoletano P. A Systematic Literature Review of Innovations, Challenges, and Future Directions in Telemonitoring and Wearable Health Technologies. IEEE Journal of Biomedical and Health Informatics. 2025;.

[4] Schmidt P, Reiss A, Duerichen R, Marberger C, Van Laerhoven K. Introducing WESAD, a multimodal dataset for wearable stress and affect detection. In: Proceedings of the 20th ACM International Conference on Multimodal Interaction; 2018. p. 400–408.

ARTICLE IN PRESS

[5] Assabumrungrat R, Sangnark S, Charoenpattarawut T, Polpakdee W, Sudhawiyangkul T, Boonchieng E, et al. Ubiquitous affective computing: A review. IEEE Sensors Journal. 2021;22(3):1867–1881.

[6] Abdelfattah E, Joshi S, Tiwari S. Machine and deep learning models for stress detection using multimodal physiological data. IEEE Access. 2025;.

[7] Liu Y, Zavareh AT, Zoghi B. Enhancing Well-Being and Alleviating Stress via Wearable-Driven Emotion Recognition and EQ Intentional Practice with Deep Reinforcement Learning. IEEE Sensors Letters. 2024;.

[8] Tian F, Zhang L, Zhu L, Zhao M, Liu J, Dong Q, et al. Advancements in affective disorder detection: Using multimodal physiological signals and neuromorphic computing based on snns. IEEE Transactions on Computational Social Systems. 2024;.

[9] Vidyasagar KC, Kumar KR, Sai GA, Ruchita M, Saikia MJ. Signal to image conversion and convolutional neural networks for physiological signal processing: A review. Ieee Access. 2024;12:66726–66764.

[10] Minor B, Greeley C, Holder R, Thomas B, Holder LB, Cook DJ. A Feature-Augmented Transformer Model to Recognize Functional Activities from in-the-wild Smartwatch Data. IEEE Journal of Biomedical and Health Informatics. 2025;.

[11] Arif S, Siddiqui MM, Alqahtani N, Shah K, Khan MA, Kraiem N. Advanced Signal Processing Algorithm-Based State Estimation and Denoising of ECG Signals for Biomedical Applications. IEEE Access. 2025;.

[12] Lee S, Lee S, Jang W, Kim J, Yoo DK, Lee J. Weighted Iterative Complex Demodulation for High-Resolution Instantaneous Frequencies in Low-Frequency PPG Signals from Wearable Devices. IEEE Journal of Biomedical and Health Informatics. 2025;.

[13] Tian Y, Wei H, Tan J. An adaptive-gain complementary filter for real-time human motion tracking with MARG sensors in free-living environments. IEEE Transactions on Neural Systems and Rehabilitation Engineering. 2012;21(2):254–264.

[14] Ardalan S, Moghadami S, Jaafari S. Motion noise cancelation in heartbeat sensing using accelerometer and adaptive filter. IEEE Embedded Systems Letters. 2015;7(4):101–104.

[15] De Sabbata G, Simonini G. Real-Time Forecasting from Wearable-Monitored Heart Rate Data Through Autoregressive Models. Journal of Healthcare Informatics Research. 2025;p. 1–20.

ARTICLE IN PRESS

[16] Tavares T, Nogueira M, Rosário D, Santos A, Cerqueira E. Traffic model based on autoregression for ppg signals in wearable networks. IEEE Networking Letters. 2020;2(2):49–53.

[17] Salem O, Alsubhi K, Mehaoua A, Boutaba R. Markov models for anomaly detection in wireless body area networks for secure health monitoring. IEEE Journal on Selected Areas in Communications. 2020;39(2):526–540.

[18] Fathian R, Khandan A, Rahmanifar N, Ho C, Rouhani H. Feasibility and validity of wearable sensors for monitoring temporal parameters in manual wheelchair propulsion. IEEE Journal of Biomedical and Health Informatics. 2024;28(9):5239–5246.

[19] Ben-Moshe N, Tsutsui K, Brimer SB, Zvuloni E, Sörnmo L, Behar JA. RawECGNet: Deep learning generalization for atrial fibrillation detection from the raw ECG. IEEE Journal of Biomedical and Health Informatics. 2024;28(9):5180–5188.

[20] Leng J, Li H, Shi W, Gao L, Lv C, Wang C, et al. Time-frequency-space EEG decoding model based on dense graph convolutional network for stroke. IEEE Journal of Biomedical and Health Informatics. 2024;28(9):5214–5226.

[21] Ljung L. System Identification: Theory for the User. 2nd ed. Upper Saddle River, NJ: Prentice Hall; 1999.

[22] Engle RF. Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation. Econometrica. 1982;50(4):987–1007.

[23] Han H, Kristensen D. Asymptotic theory for the QMLE in GARCH-X models with stationary and nonstationary covariates. Journal of business & economic statistics. 2014;32(3):416–429.

[24] Tse YK. A test for constant correlations in a multivariate GARCH model. Journal of econometrics. 2000;98(1):107–127.

[25] Yu C, Li D, Jiang F, Zhu K. Matrix GARCH model: Inference and application. Journal of the American Statistical Association. 2025;120(551):1747–1762.

[26] Zhu Q, Tan S, Zheng Y, Li G. Quantile autoregressive conditional heteroscedasticity. Journal of the Royal Statistical Society Series B: Statistical Methodology. 2023;85(4):1099–1127.

[27] Blundell R, Bond S. Initial conditions and moment restrictions in dynamic panel data models. Journal of econometrics. 1998;87(1):115–143.

[28] Tibshirani R. Regression Shrinkage and Selection via the Lasso. Journal of the Royal Statistical Society: Series B (Methodological). 1996;58(1):267–288.

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[29] Zou H. The Adaptive Lasso and Its Oracle Properties. Journal of the American Statistical Association. 2006;101(476):1418–1429.

[30] Wang H, Li G, Tsai CL. Regression coefficient and autoregressive order shrinkage and selection via the lasso. Journal of the Royal Statistical Society Series B: Statistical Methodology. 2007;69(1):63–78.

[31] Medeiros MC, Mendes EF. L1-Regularization of High-Dimensional Time-Series Models with Non-Gaussian and Heteroskedastic Innovations. Journal of Econometrics. 2016;191(1):150–164.

[32] Liu S, Lu M. Variance-Guided Regression for Heteroscedastic Data with a Grouping-Based Extension for Nonlinear Prediction. Statistics in Medicine. 2026;45(13-14):e70632. https://doi.org/10.1002/sim.70632.

[33] UC Irvine Machine Learning Repository.: WESAD (Wearable Stress and Affect Detection). Accessed 14 May 2026. Available from: https://archive.ics.uci.edu/ml/datasets/WESAD+%28Wearable+Stress+and+Affect+Detection%29.

[34] MacNab Y, Dean C. Parametric bootstrap and penalized quasi-likelihood inference in conditional autoregressive models. Statistics in medicine. 2000;19(17-18):2421–2435.

[35] Fan J, Li R. Variable Selection via Nonconcave Penalized Likelihood and Its Oracle Properties. Journal of the American Statistical Association. 2001;96(456):1348–1360. https://doi.org/10.1198/016214501753382273.

[36] Zhang CH. Nearly Unbiased Variable Selection under Minimax Concave Penalty. The Annals of Statistics. 2010;38(2):894–942. https://doi.org/10.1214/09-AOS729.

[37] Makowski D, Pham T, Lau ZJ, Brammer JC, Lespinasse F, Pham H, et al. NeuroKit2: A Python toolbox for neurophysiological signal processing. Behavior research methods. 2021;53(4):1689–1696.

[38] R Core Team.: R: A Language and Environment for Statistical Computing. Vienna, Austria. Available from: https://www.R-project.org/.

[39] Liaw A, Wiener M. Classification and Regression by randomForest. R News. 2002;2(3):18–22.

[40] Ridgeway G, GBM Developers.: gbm: Generalized Boosted Regression Models. R package version 2.2.2. Available from: https://CRAN.R-project.org/package=gbm.

[41] Lu M, Yin R, Chen XS. Ensemble Methods of Rank-Based Trees for Single Sample Classification with Gene Expression Profiles. Journal of Translational Medicine.

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2024;22(140). https://doi.org/10.1186/s12967-024-04940-2.

[42] Wang Z, Wang Y, Hu C, Yin Z, Song Y. Transformers for EEG-based emotion recognition: A hierarchical spatial information learning model. IEEE Sensors Journal. 2022;22(5):4359–4368.

[43] Han Z, Zhao J, Leung H, Ma KF, Wang W. A review of deep learning models for time series prediction. IEEE Sensors Journal. 2019;21(6):7833–7848.

[44] McGonigle ET, Cho H. Nonparametric data segmentation in multivariate time series via joint characteristic functions. Biometrika. 2025;112(2):asaf024.

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Response to Reviewers

Manuscript JTRM-D-26-10410: "Variance-Aware Penalized Panel Models for Temporal Risk Detection from Wearable Sensor Data"

We thank the editor and the four reviewers for their careful and constructive reading of the manuscript. Below we respond point-by-point to each comment. Text in light blue summarizes how the original manuscript addressed the point; text in navy blue describes the clarifications, new analyses, or corrections made in the revised manuscript. All page/section references are to the revised manuscript and the expanded Supplementary Material (now eight sections, S1–S8).

Reviewer #1

Reviewer comment. *This manuscript proposes a variance-aware penalized panel ARX-GARCHX model for wearable sensor data. Its goal is to transform noisy, high-dimensional physiological streams into an interpretable, subject-specific temporal risk score. The model separates conditional mean dynamics from conditional variance dynamics, uses sparse penalties to handle many multimodal covariates, and estimates the probability that a subject's heart rate exceeds an individualized threshold.*

The manuscript evaluates the approach through simulations and a WESAD wearable-stress demonstration. In simulations, variance-aware models perform better when volatility is structured by seasonality or regime changes. In WESAD, the proposed risk score achieves an AUC of 0.931, improving over raw heart-rate mean with AUC 0.868, and performs competitively with supervised classifiers.

Overall, the manuscript's principal strength lies in its interpretable and methodologically grounded approach to wearable-sensor risk modeling. By explicitly modeling both mean dynamics and conditional variance, the proposed framework addresses an important limitation of many existing wearable-sensing pipelines that treat variability primarily as noise. The subject-adaptive thresholding strategy and probability-scale risk score are particularly appealing for translational applications, where interpretability and individual baseline differences are important. The combination of simulation studies and a real-data WESAD demonstration further strengthens the manuscript by showing the potential utility of the approach under both controlled synthetic settings and a recognized wearable-stress benchmark. While additional validation is needed, especially in larger and prospective cohorts, the manuscript offers a promising and clearly motivated alternative to purely black-box classification methods.

We thank Reviewer 1 for the careful reading and encouraging assessment. We are glad the reviewer found the explicit separation of conditional mean and variance dynamics, the subject-adaptive thresholding, and the probability-scale risk score to be well suited to translational use, and that pairing controlled simulations with the WESAD benchmark strengthens the evidence for the approach. The reviewer requested no specific revisions, but raised one substantive point—that additional validation is needed, especially in larger and prospective cohorts—which we take seriously and address directly here.

We fully agree, and three changes made during the revision speak to this point. (i) The WESAD

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comparison is now reported with participant-clustered bootstrap confidence intervals that resample whole participants, so the evidence reflects the modest number of individuals (n=15); the score's advantage over the raw heart-rate baseline survives this clustering-aware inference (paired-AUC difference CI $[0.031, 0.094]$, excluding zero). (ii) A new leak-free threshold analysis (Supplementary Section S6) re-estimates the exceedance threshold from an early per-participant burn-in window—mimicking how the score would be fixed prospectively from a short calibration period rather than from the full recording—and discrimination remains at or above the raw baseline; we accordingly recommend this calibration-based thresholding for prospective use. (iii) We retain and sharpen the proof-of-concept framing in Section 5, where validation in larger, longitudinal, and prospectively labeled cohorts is identified as the essential next step.

Reviewer #2

Reviewer comment. *Considering the heteroscedasticity of wearable physiological signals (heart rate, EDA, etc.), this manuscript proposes a penalized panel ARX–GARCHX model that jointly fits conditional mean and variance dynamics, with Lasso penalties to handle high-dimensional multimodal covariates in both equations. The output is a subject-specific exceedance probability: the estimated chance the target signal crosses a personalized threshold given recent history and covariates. Simulations across four synthetic regimes show that explicitly modeling covariate-driven variance can reduce risk-score RMSE when volatility is structured. On the WESAD stress benchmark, the proposed method is comparable to supervised methods which have access to the true label but ignore the temporal dependence.*

Major comments

Major 1.

Reviewer comment. *As discussed in Sec 5, the risk score depends on the threshold, and so does the interpretation based on the proposed risk score. I would suggest highlighting this in Sec 2.4 where the risk score is proposed.*

Section 2.4 defined the exceedance score $\pi_{s,t}(c_s)$ and noted that choosing c_s as a participant-specific quantile makes the score comparable across participants, with threshold dependence discussed in Section 5. We now make the threshold dependence explicit at the point of definition in Section 2.4, stating that the score and its interpretation are conditional on c_s, that a different c_s defines a different exceedance event, and that the threshold should be chosen to match the intended use and reported alongside the score, with a forward pointer to Sections 4.2 and 5.

Manuscript revision. "The score and its interpretation are explicitly conditional on the choice of threshold c_s: $\pi_{s,t}(c_s)$ answers 'how likely is the signal to exceed c_s right now,' so a different c_s defines a different exceedance event and a different score. The threshold should therefore be chosen to match the intended use—for example a within-subject quantile for baseline-relative monitoring, or a clinically anchored value when one is available—and reported alongside the score."

Major 2.

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Reviewer comment. *Related to the point above, the threshold c_s in the WESAD study seems to be chosen as the quantile across the entire timeframe. I'm concerned if there is any information leak there. A possible alternative is to choose the threshold on some hold-out data or the burn-in data.*

The main analysis set $c_s(0.70)$ as the 70th percentile of each participant's `HR_mean` over the full recording. We agree this deserves scrutiny. Although c_s uses no stress labels, it does include the entire recording, including the windows later used for evaluation, so it can lend a modest optimism to the reported discrimination. We added a sensitivity analysis (new Supplementary Section S6) that re-estimates c_s *without* such information—from an early per-participant burn-in window (first 20%/30% of windows) and from baseline-annotated windows only—while holding the fitted mean and variance fixed. The leak-free burn-in threshold reduces AUC_{ex} from 0.936 to approximately 0.88, which remains at or above the raw `HR_mean` baseline (0.870). We report this in Section 4.3 and temper the claim in Section 5, and recommend fixing c_s prospectively from a calibration period.

Manuscript revision. New Supplementary Section S6 (Table S4, Fig. S1). Section 4.3: "First, replacing the full-recording threshold $c_s(0.70)$ with a leak-free threshold estimated solely from an early burn-in window reduces AUC_{ex} from 0.931 to approximately 0.880; notably, this remains at or above the raw `HR_mean` baseline (details in Supplementary Section S6)." Section 5: 'Because this quantile is computed over the entire recording, it inherently incorporates the same elevated heart-rate windows that later define the positive class. Although no labels are used, this may introduce a modest optimistic bias to the reported discrimination. A sensitivity analysis fixing c_s from a leak-free early burn-in window (Supplementary Section S6) reduces AUC_{ex} from 0.931 to approximately 0.880 but remains at or above the raw heart-rate baseline. This confirms that the temporal signal is genuine while emphasizing that the exact magnitude of performance is sensitive to threshold selection. In future translational applications, thresholds should be fixed prospectively from an initial calibration period and chosen from clinical anchors, patient-reported outcomes, or adverse-event definitions, rather than relying solely on empirical quantiles computed over the full recording."

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Major 3.

Reviewer comment. *Simulations in Sec 3 compare the proposed ARX-GARCHX with three other variants in the ARX-GARCH family. I'm wondering why ARX_p-GARCHX is not included (penalized mean, X-dependent variance). Moreover, I would suggest adding as a baseline AR-GARCH which doesn't include any covariates.*

The original study compared four variants (Models A–D) differing in whether the mean/variance are penalized and whether X enters the variance. We added both requested models to Section 3 and Supplementary Table S1: **Model E** (ARX_p–GARCHX; penalized mean, *unpenalized* X-dependent variance), which isolates the effect of penalizing the variance covariates (E vs. D), and **Model F** (AR–GARCH; no covariates in the mean or variance), a baseline that quantifies the value added by covariates. Because Models A–D have already been estimated over 100 replications and reported in Table S1, we compute *only* E and F, using the same generator, evaluator, tuning ($\lambda_\beta = 0.01$ for the penalized mean), and seed scheme as the published A–D run, so that A–D remain unchanged

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and the two new columns are directly comparable. The completed 100-replication run confirms the anticipated pattern: A–D reproduce the published Table S1 values exactly; E is essentially indistinguishable from D (and is numerically the most accurate model at T=240, with the lowest RMSE in all four scenarios); and F has by far the highest RMSE in every scenario ($\approx$ 0.10–0.16 versus 0.05–0.09 for the covariate-driven models), quantifying the accuracy lost by ignoring covariates entirely.

Manuscript revision. Section 3.2 model list expanded to Models A–F. Supplementary Table S1 now reports columns E and F alongside the unchanged A–D (A–D reproduce the published values exactly), and Figure 3 is redrawn with six models. In Section 3.3, we added "Model **E** (ARX$_p$–GARCHX; penalized mean, unpenalized X-dependent variance) is essentially indistinguishable from Model **D**. Indeed, Model **E** is the most accurate model at the longer horizon of $T = 240$, achieving the lowest RMSE across all four scenarios; this indicates that penalizing the variance covariates primarily stabilizes estimation at small T rather than shifting the central tendency. At the other extreme, the covariate-free baseline **F** (AR–GARCH) consistently yields the highest RMSE in every scenario (roughly 0.10–0.16 versus 0.05–0.09 for the covariate-driven models), quantifying the accuracy lost by ignoring exogenous covariates altogether. Taken together, these six models bracket the value of covariate-driven variance: it is decisive when volatility is structured, negligible when it is not, and best realized by a penalized specification that couples covariate information with regularized estimation. Full results for all six models are reported in Supplementary Table S1."

Major 4.

Reviewer comment. *Sec 3.3, second paragraph: it says that the bias of Model C (and D) can be due to the shrinkage in the variance equation, but Model C is ARX-GARCHX, which doesn't apply any penalty.*

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The original sentence attributed the mild positive bias of both C and D to "shrinkage in the variance equation." The original wording was incorrect for Model C: it applies no penalty in the variance equation, so "shrinkage" cannot be its cause. We corrected the passage to distinguish the two mechanisms. In Model D the bias reflects ℓ_1 shrinkage of the variance-covariate coefficients γ; in the unpenalized Model C it instead arises from finite-sample estimation noise in the high-dimensional two-stage variance fit (many near-zero coefficients estimated without regularization) together with the positive-variance safeguard, which inflate small fitted variances. Both effects are small.

Manuscript revision. "In Model **D**, which applies an ℓ_1 penalty to the variance-covariate coefficients γ, the bias reflects shrinkage of γ toward zero, which mildly attenuates the fitted conditional variance and shifts exceedance probabilities upward through the curvature of $z \mapsto 1 - \Phi(z)$. Model **C**, by contrast, applies *no* penalty in the variance equation; its bias instead arises from finite-sample estimation noise when the high-dimensional variance-covariate vector is fit without regularization (many near-zero coefficients estimated from short series), together with the positive-variance safeguard, which together inflate small fitted variances."

Minor comments

Minor 1.

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Reviewer comment. *`HR_mean` used in the y-axis label and legend Figure 1 is not defined by Figure 1.*

We now define it where Figure 1 is introduced: "`HR_mean` denotes the mean heart rate (in beats per minute) within each 60-s window, the target signal used throughout."

Minor 2.

Reviewer comment. *Sec 2.2, first paragraph, second sentence: I would probably consider using either ℓ_1 penalty or LASSO penalty but not both.*

Revised to "In this paper, we employ the ℓ_1 penalty to select informative covariates in both the conditional mean and conditional variance equations."

Minor 3.

Reviewer comment. *Section 3, Table 1: any reason why burn-in is different across the four scenarios in the simulation study?*

Table 1 listed burn-ins of 200/220/180/180 for S1–S4 without explanation.

Thank you for pointing this out. The burn-in period is used solely during data generation to remove initialization transients before retaining observations for analysis. It is discarded prior to model fitting and therefore has no role in the estimation procedure or the quantities being evaluated. The small differences in burn-in length across scenarios were chosen only for simulation convenience to ensure that the retained observations are approximately from the stationary regime. We have added a clarifying note to Table 1.

Manuscript revision. Table 1 note: "The burn-in period is used only during data generation to remove initialization transients and is discarded before model fitting. Consequently, it does not affect the estimation procedure, the estimands, or the reported Bias and RMSE. The modest differences in burn-in lengths across scenarios were chosen for simulation convenience to ensure that the retained observations are approximately from the stationary regime."

Minor 4.

Reviewer comment. *The supervised comparators in the WESAD study ignore temporal dependence. It would be interesting to know how the supervised comparators perform if they are used to model the residuals of AR-GARCH, though the improvement may be limited since the AUC is already high with the temporal dependence ignored.*

The supervised comparators (logistic, RF, GBM) were trained on the raw 77 window features under LOSO. We thank the reviewer for this helpful suggestion. We performed the requested analysis by first fitting a covariate-free panel AR–GARCH model to `HR_mean`, computing the standardized residual $z_{s,t} = (y_{s,t} - \widehat{\mu}_{s,t})/\widehat{\sigma}_{s,t}$, and then repeating the same LOSO supervised analyses after augmenting the original feature set with $z_{s,t}$. Consistent with the reviewer's expectation, the results changed very little: logistic regression AUC changed from 0.931 to 0.930, random forest from 0.928

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to 0.924, and gradient boosting from 0.945 to 0.946. These results suggest that, for this dataset, the supervised classifiers already capture most of the discriminative information, so explicitly removing temporal dependence through an AR–GARCH preprocessing step provides little additional benefit. Because this analysis is supplementary to the main contribution of the paper, we have made the corresponding scripts and outputs available in the GitHub reproducibility repository rather than expanding the manuscript.

Reviewer #3

Reviewer comment. *In this manuscript, the authors develop a variance-aware penalized panel ARX-GARCHX model for multimodal wearable-sensor data. The framework combines subject-specific mean and variance baselines, shared autoregressive dynamics, sparse covariate effects, and a conditional exceedance probability as a normalized temporal score. The method is evaluated in simulations and illustrated using 60-second chest-sensor windows from the 15-participant WESAD dataset.*

The manuscript addresses a relevant problem in translational digital-biomarker research. Explicitly separating conditional mean and variance dynamics is a useful contribution, the simulations are well aligned with the methodological objective, and the WESAD analysis is appropriately framed as a proof-of-concept rather than clinical validation. Overall, the paper is clearly written and potentially suitable for Journal of Translational Medicine. The remaining issues are limited and can be addressed through clarification, modest supplementary analyses, and tempering of several claims. I have the following comments.

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Comment 1.

Reviewer comment. *Please clarify the WESAD endpoint and evaluation in Section 4.3: define the positive and negative classes, explain how transition/other windows were handled, and report window and class counts by participant. Because repeated windows are clustered within only 15 participants, the pooled DeLong p value may overstate precision. Participant-clustered bootstrap confidence intervals would be preferable, or the comparison should be presented as descriptive. The supervised classifiers use leave-one-subject-out cross-validation, whereas the proposed score appears to be fitted retrospectively to the full recordings. Unless the score is evaluated under the same held-out design, please describe the supervised AUCs as contextual benchmarks rather than a formal head-to-head comparison.*

Section 4.3 evaluated the exceedance score against annotated WESAD segments and reported a pooled DeLong test, with supervised LOSO classifiers as reference. Thank you for this helpful comment. (i) **Evaluation endpoint**: the positive class is the set of *stress*-annotated windows (WESAD label 2), whereas the negative class consists of all other retained windows (baseline, amusement, and transition/other); transition and amusement windows are therefore retained rather than discarded. (ii) **Window counts**: after preprocessing, 1,427 windows from 15 participants remained (182 stress and 1,245 non-stress), and the per-participant window and class counts are now reported in new Supplementary Table S3. (iii) **Clustered inference**: because repeated windows are clustered within participants, we now additionally report participant-clustered bootstrap confidence intervals obtained by resampling whole participants (3,000 bootstrap replicates). The resulting 95% confidence intervals are $[0.887, 0.972]$ for the proposed risk score and $[0.821, 0.918]$ for `HR_mean`; the paired AUC difference has a 95% confidence interval of $[0.031, 0.094]$, which excludes zero. (iv)

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Comparison with supervised methods: we now explicitly state that the proposed score is evaluated in-sample, whereas the supervised classifiers are evaluated under leave-one-subject-out cross-validation. Accordingly, the supervised AUCs are presented as *contextual benchmarks* rather than a formal head-to-head comparison, and the intended primary comparison is against the raw `HR_mean` baseline, which is evaluated under the same design.

Manuscript revision. New paragraph in Section 4.2 ("Evaluation endpoint"):

"For reference-based evaluation, we define a binary window-level endpoint. The positive class consists of windows annotated as *stress* (WESAD label 2), whereas the negative class pools all other retained windows (baseline, amusement, and transition/other). Transition and amusement windows are therefore treated as non-stress rather than discarded. After preprocessing, 1,427 windows from 15 participants were available (182 stress and 1,245 non-stress); the corresponding per-participant window and class counts are reported in Supplementary Table S3. The annotations are used only for evaluation and never for model fitting."

The final paragraph of Section 4.2 has also been revised as follows:

"Because the proposed score and the supervised classifiers are not evaluated under a common held-out design, the supervised AUCs are presented as *contextual benchmarks* that illustrate the performance of purely discriminative window-level methods, rather than as a formal head-to-head comparison. The intended primary comparison for the proposed score is against the raw `HR_mean` baseline, which is evaluated under the same experimental design."

Section 4.3 now additionally states:

"Because repeated windows are clustered within only 15 participants, the pooled DeLong test may overstate precision. We therefore additionally report participant-clustered bootstrap confidence intervals obtained by resampling whole participants (3,000 bootstrap replicates). This yields 95% confidence intervals of [0.887, 0.972] for $\mathrm{AUC}_{\mathrm{ex}}$ and [0.821, 0.918] for $\mathrm{AUC}_{\mathrm{HR}}$, while the paired AUC difference has a 95% confidence interval of [0.031, 0.094], excluding zero."

New Supplementary Section S5 (Table S3) reports the per-participant window counts, class counts, and details of the participant-clustered bootstrap procedure.

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Comment 2.

Reviewer comment. *The score is the conditional probability that HR_mean exceeds an individualized threshold, not the probability of stress or a clinical event. Please align the terminology in the Abstract, Results, Discussion, and Figure 4, retaining wording such as "stress-associated windows". Please also state clearly that the WESAD threshold, scaling, and subject-specific parameters are estimated from the analyzed recordings. A brief explanation of how an initial calibration period would be used prospectively, and whether contemporaneous covariates make the score an end-of-window representation or a one-step-ahead forecast, would be helpful.*

The manuscript already described the score as an "exceedance-based" risk score and used "stress-associated windows" in places, but some passages could be read as estimating a probability of stress. Thank you for this helpful comment. We have aligned the terminology throughout the manuscript (Abstract, Results, Discussion, and Figure 4) so that the proposed score is consistently described as the estimated conditional probability that `HR_mean` exceeds the individualized threshold c_s, while

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retaining "stress-associated windows" to refer to the WESAD annotations. We also added an explicit clarification in Section 4.2 that (i) the score is *not* a probability of stress or any clinical event; (ii) because the variance model uses contemporaneous covariates (M=0), the proposed score is an end-of-window representation rather than a one-step-ahead forecast, whereas a forecasting version can be obtained by restricting both the mean and variance models to strictly lagged covariates; and (iii) the threshold, within-subject covariate standardization, and subject-specific parameters (α_s, ω_s) are all derived from the analyzed recordings. We further clarify that, in a prospective deployment, these quantities would instead be obtained during an initial subject-specific calibration period and then held fixed during subsequent monitoring.

Manuscript revision. New paragraph in Section 4.2:

"To clarify the interpretation and estimation of the proposed score, $\widehat{\pi}_{s,t}(c_s)$ denotes the estimated conditional probability that `HR_mean` exceeds the individualized threshold c_s; it is *not* a probability of stress or of any clinical event. Because the conditional variance model uses contemporaneous covariates $x_{s,t}$ (lag order M=0), $\widehat{\pi}_{s,t}$ represents the current window rather than a one-step-ahead forecast. A forecasting variant can be obtained by restricting both the mean and variance equations to strictly lagged covariates. In this demonstration, the threshold c_s, the within-subject covariate standardization, and the subject-specific parameters (α_s, ω_s) are all derived from the analyzed recordings. In a prospective deployment, these quantities would instead be obtained during an initial subject-specific calibration period and then held fixed during subsequent monitoring. Specifically, $c_s(0.70)$ is defined as the 70th percentile of each subject's `HR_mean` distribution over the full recording. It is therefore computed without using any stress annotations, although it is estimated from the entire recording, including the windows that are later used for evaluation. A sensitivity analysis that instead estimates c_s from a leak-free calibration period is reported in Supplementary Section S6 and summarized in Section 4.3. "

The terminology in the Abstract, Results, and Discussion has been revised accordingly so that the proposed quantity is consistently described as an exceedance-based risk score—that is, the estimated conditional probability that `HR_mean` exceeds an individualized threshold—rather than as a probability of stress or of a clinical event. Figure 4 has been updated only to improve wording for consistency with this terminology.

Comment 3.

Reviewer comment. *To link the real-data results more directly to the central contribution, please consider adding one focused WESAD ablation comparing the full penalized ARX-GARCHX model with an otherwise comparable model that excludes covariates from the variance equation, for example Model B versus Model D. Reporting the corresponding AUC and ideally a simple calibration measure for the actual HR-threshold exceedance event would be sufficient. This analysis can be placed in the Supplementary Material.*

The main text reported only the full model's discrimination. Thank you for this helpful suggestion. We have added the requested ablation as new Supplementary Section S7. Model B (ARX_p–GARCH, without covariates in the variance equation) and Model D (the full ARX_p–GARCHX_p model) share the same penalized mean specification and differ only in whether covariates enter the conditional variance equation. We report discrimination for both the annotated stress endpoint and the actual

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heart-rate threshold-exceedance event, together with calibration for the threshold-exceedance event using the Brier score. Model D achieved an AUC of 0.921 for the annotated stress endpoint and 0.927 for the heart-rate threshold-exceedance event, with a Brier score of 0.085. Model B achieved corresponding values of 0.934, 0.929, and 0.084, respectively. Both substantially outperformed the raw `HR_mean` baseline (AUC 0.868), whereas the differences between Models B and D were small. These results complement the simulation study by showing that the proposed framework is robust to whether covariates are included in the variance equation for the WESAD dataset, while the simulation study demonstrates the benefit of covariate-driven variance modeling under structured heteroscedasticity.

Manuscript revision. New Supplementary Section S7 (Table S5) reports the requested WESAD ablation. Model B (ARX$_p$–GARCH) and Model D (ARX$_p$–GARCHX$_p$) are compared under identical settings using both the annotated stress endpoint and the exceedance event $\{y_{s,t} > c_s(0.70)\}$. Discrimination (AUC) and calibration (Brier score) are reported for both models together with the raw `HR_mean` baseline.

Comment 4.

Reviewer comment. *Please expand the reproducibility details for the WESAD analysis by providing the complete list of 77 features, retained windows per participant, handling of missing samples or failed feature calculations, and how the balanced-panel assumption was reconciled with differing recording lengths or quality-control exclusions. Please also report the tuning grid and temporal fold construction, software/package versions, and the exact code release or commit used for the reported results.*

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The manuscript summarized feature blocks (Table 2) and stated the lag orders and window rule. Thank you for this helpful suggestion. We have substantially expanded the reproducibility documentation for the WESAD analysis. The main manuscript now clarifies the feature-selection procedure (including correlation-based pruning to the final 77 features), the retained window counts per participant, the quality-control and missing-data handling rules, and explains that the analyzed data form an unbalanced panel, whereas the balanced-panel notation in Section 2.1 is used only for expositional simplicity because all lagged quantities are constructed separately within each participant. New Supplementary Section S8 provides the complete list of the 77 retained features, additional preprocessing details, the tuning grid and blockwise time-series cross-validation scheme, software and package versions, and the code used to generate the reported results. For the review process, the manuscript-specific reproducibility scripts and outputs are provided in the `manuscript/` subdirectory of the public repository.

Manuscript revision. Section 4.1 has been expanded to describe the correlation-based pruning of the candidate feature set to the final 77 retained features, the retained window counts for each participant, the quality-control and missing-data handling procedures, and the reconciliation between the balanced-panel notation and the unbalanced WESAD data. New Supplementary Section S8 provides the complete list of the 77 retained features (Table S6), per-participant window counts, additional preprocessing details, the tuning grid, blockwise time-series cross-validation scheme, software and package versions (R 4.5.3; `glmnet` 5.0; `pROC` 1.19.0.1; `randomForest` 4.7.1.2; `gbm` 2.2.3; `caret` 7.0.1; Python with `NeuroKit2`), together with the software and reproducibility resources referenced in the Code Availability

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statement. Specifically, the `varGuidTS` software package is available through CRAN, while the complete manuscript-specific reproducibility scripts and outputs are available in the public GitHub repository under https://github.com/zionwzz/variance-guided-risk-demo/tree/main/manuscript.

Comment 5.

Reviewer comment. *Several presentation points can be handled editorially. In Section 2.2, revise the sentence beginning "In this paper, we focus on the ℓ_1 penalty" and define the $ARX_p/GARCHX_p$ notation at first use. In Figure 1, please identify arrhythmia and sleep apnea as hypothetical applications and remove wording such as "supplied for this revision". In Figure 4, please identify the representative participant and state how that trajectory was selected.*

Thank you for these helpful suggestions. We have revised the manuscript accordingly. Section 2.2 now uses consistent terminology for the ℓ_1 penalty, and the $\mathrm{ARX}_p/\mathrm{GARCHX}_p$ notation is defined at its first use in Section 3.2. The Figure 1 caption now explicitly identifies arrhythmia and sleep apnea as hypothetical illustrative applications, and the wording "supplied for this revision" has been removed. For Figure 4, the representative participant is now identified as WESAD subject S11, and the text explains that this participant was selected before examining the model output as a representative case with a clearly delineated stress segment.

Manuscript revision. Section 2.2 has been revised to use consistent ℓ_1-penalty terminology, and the $\mathrm{ARX}_p/\mathrm{GARCHX}_p$ notation is now defined at its first use in Section 3.2. The Figure 1 caption now explicitly states that arrhythmia and sleep apnea are hypothetical illustrative applications and removes the phrase "supplied for this revision." The corresponding text for Figure 4 now identifies the representative participant as WESAD subject S11 and explains that this trajectory was selected before examining the model output as a representative case with a clearly delineated stress segment.

Reviewer #4

Reviewer comment. *Thank you for the opportunity to review this manuscript! The study presents a well-conducted and clearly motivated methodological contribution for temporal risk detection from wearable sensor data. The statistical methodology is appropriate, the manuscript is generally well organized, and the simulations and real-data application are presented clearly. Overall, I found the work to be technically sound and the conclusions to be supported by the analyses presented.*

I have only a few minor comments, primarily related to language and presentation, which I believe could further improve the readability of the manuscript.

Comments 1–2.

Reviewer comment. *(1) On page 2, lines 19–21, currently stated as "... or feature extraction that does not explicitly separate changes in location from changes in volatility. This distinction is important in...", consider revising as "... or feature extraction without explicitly separating changes in location from changes in volatility. Distinguishing changes in location from changes*

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in volatility is important. . . ". (2) On page 3, lines 1–6, "In many sensing applications, the goal extends beyond forecasting to . . . Wearable streams often show . . . motivating. . . ", consider moving these two sentences to the end of the third paragraph on page 1 to improve the flow of the text.

As a general suggestion, although the manuscript is already well written, I encourage the authors to perform a final round of language polishing throughout the manuscript to further improve the flow, readability, and overall presentation.

We thank the reviewer for the encouraging assessment of our work and for these helpful editorial suggestions. We adopted suggestion (1) verbatim and implemented suggestion (2) by moving the two sentences beginning "In many sensing applications, the goal extends beyond forecasting . . . " to the end of the third paragraph of the Introduction, improving the overall flow of the discussion. We also performed a final language-editing pass throughout the manuscript to improve readability, consistency, and presentation, including minor wording refinements and typographical corrections.

Manuscript revision. Suggestion (1) was adopted verbatim in Section 1; the passage now reads: "These approaches demonstrate the value of continuous physiological monitoring, but many rely on dense labels, short-window classification, or feature extraction without explicitly separating changes in location from changes in volatility. Distinguishing changes in location from changes in volatility is important in biomedical and translational studies, where a risk-relevant physiological deviation may depend on a participant's own baseline and variance pattern rather than on a fixed population threshold." For suggestion (2), the two sentences "In many sensing applications, the goal extends beyond forecasting to deriving a *temporal risk score* that captures mean dynamics and variance fluctuations under realistic noise. Wearable streams often show sharp volatility changes driven by motion or context, motivating explicit variance modeling when extracting stable temporal representations from noisy signals." now appear at the end of the third paragraph of Section 1.

We thank all four reviewers again for their thoughtful and constructive comments. The revisions have strengthened the manuscript's scientific rigor through participant-clustered inference, a leak-free threshold sensitivity analysis, an expanded six-model simulation comparison, and a focused WESAD variance-covariate ablation, while substantially improving the reproducibility and transparency of the analyses. We believe the revised manuscript more clearly communicates the proposed methodology while maintaining its intended proof-of-concept translational framing.